

Quantitative Analysis of 8-Bit Integer Quantization on EfficientNet Lite for Edge Computing: 9.5x Latency Reduction in Multi-Domain Medical Diagnostics

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Abstract—This work explores methods for enabling medical image analysis models to function effectively on platforms where processing capacity is tightly limited. While advanced Convolutional Neural Networks (CNNs) offer high diagnostic potential across multi-domain pathologies (skin lesions, oral diseases, and nail disorders), their deployment is inherently limited by large model size and high inference latency.

To overcome this, we systematically investigate model efficiency by benchmarking three distinct optimization tiers—Full-Precision (FP32), Half-Precision (FP16), and Fully Quantized (8-bit Integer, INT8)—derived from the EfficientNet Lite architecture. Our methodology focuses on quantifying the trade-off between diagnostic accuracy, model size, and execution speed using a rigorous comparative evaluation protocol.

The results show that the 8-bit quantized model offers the most practical balance for deployment. Reducing the original FP32 model from roughly 46 MB to 4.69 MB gives a tenfold drop in size, and the inference time improves by about eleven times, averaging 17 ms per image. The test accuracy remained low (around 25%), which reflects the baseline model’s limited ability to generalize rather than a failure of the quantization process. Even so, the INT8 version delivers real-time performance with a very small memory footprint across all three clinical domains.

This study provides a useful reference point for future work on edge-AI systems. It shows that full integer quantization is a practical way to achieve fast, stable, and portable inference on mobile diagnostic devices.

Index Terms—Model Quantization, Edge Computing, EfficientNet Lite, Medical Image Analysis, Model Compression

INTRODUCTION

Deep Learning (DL) has achieved remarkable accuracy levels in medical image analysis, particularly in complex classification tasks such as dermatology, pathology, and radiology [1, 2]. These advancements promise a significant leap towards accessible and rapid clinical decision support. However, the models achieving state-of-the-art performance, like the advanced Convolutional Neural Networks (CNNs), are often computationally prohibitive. Their large memory footprints and high latency fundamentally restrict their deployment to powerful cloud servers, which contradicts the growing need for real-time, accessible, and privacy-preserving diagnostics at the network edge (i.e., on mobile devices and embedded systems).

The tension between achieving high diagnostic accuracy and ensuring efficient real-world deployment lies at the heart of

this study. To close this gap, optimizing the model becomes essential. Among the available methods, quantization offers one of the most practical ways to shrink model size and boost inference speed, while maintaining almost the same level of accuracy [3]. In simple terms, quantization converts model weights and activations from high-precision floating-point values into lower-bit fixed-point formats.

In this work, we explore which quantization strategy delivers the best balance between performance and efficiency for use in resource-limited environments. Our focus spans three important areas of dermatological image analysis: skin lesions, oral diseases, and nail disorders. For each category, we compare three optimization levels — the Full-Precision (32-bit Floating Point, FP32) baseline, the Half-Precision (16-bit Floating Point, FP16) version, and the Fully Quantized (8-bit Integer, INT8) model.

The primary scientific contribution of this work is the comprehensive empirical analysis of the Speed-Size-Accuracy trade-off across these three optimization tiers applied to multi-domain medical image classification tasks. By meticulously measuring and comparing the resultant model size, diagnostic accuracy, and, most critically, on-device inference time, this research aims to:

1. Quantify the exact gains in speed and size achieved by FP16 and INT8 quantization relative to the FP32 baseline.
2. Determine the practical impact of each quantization level on diagnostic accuracy across three independent clinical scenarios.
3. Identify the optimal, robust quantization strategy that maximizes deployment efficiency (minimal size and maximal speed) while preserving the necessary clinical reliability across diverse medical domains.

The remainder of this paper is organized as follows: Section 2 reviews related work in lightweight deep learning and model quantization. Section 3 explains the materials, model architecture, and rigorous evaluation methodology. Section 4 presents the comparative analysis results and discussion. Finally, Section 5 concludes the study and suggests future research directions.

RELATED WORK

The successful deployment of deep learning models on real-world edge devices depends largely on overcoming practical limits of memory, power, and latency. Previous studies have approached this challenge from two main directions: designing more computationally efficient network architectures, and applying strong model compression techniques to make existing models lighter and faster.

Lightweight Architectures for Medical Image Analysis

Because large, high-capacity CNNs are difficult to run efficiently on mobile or embedded hardware, researchers began developing lighter network designs that trade a small amount of accuracy for major gains in speed and memory use. Many of these newer models rely on methods such as depthwise separable convolutions, which sharply cut down the number of parameters and floating-point operations (FLOPs) needed during inference [4, 5].

The EfficientNet family — especially the Lite variants — has become a leading choice for combining accuracy with efficiency, making it well suited for TensorFlow Lite and mobile deployment. While prior work using these architectures has shown strong diagnostic performance in fields like pathology and dermatology, most studies have mainly emphasized accuracy outcomes, giving less attention to detailed runtime performance analyses that are essential for real-world deployment under limited hardware conditions.

Model Quantization and Efficiency Analysis

Quantization is one of the most powerful ways to improve the efficiency of model deployment. It works by reducing the bit-width of a network’s parameters and activations, which in turn cuts down model size and speeds up inference. This gain comes from using the fast, integer-based computation units that are already built into most modern processors [6].

Quantization approaches are generally grouped by their target precision:

- **16-bit Floating Point (FP16) Quantization:** This reduces the original 32-bit weights to 16-bit. The model becomes about half its original size, while accuracy barely changes. FP16 is usually used as an early optimization step, providing noticeable efficiency gains without compromising clinical performance [7].
- **8-bit Integer (INT8) Quantization:** This method changes the original 32-bit weights to 8-bit, usually using Post-Training Quantization (PTQ). It makes the model much smaller—up to four times smaller—and allows it to run faster. Since it uses fixed-point numbers, there can be minor rounding errors that slightly affect accuracy. That’s why it’s important to test carefully, especially when working with medical data [8].

Although many studies compare standard FP32 models to their INT8 counterparts on general-purpose tasks, comprehensive, cross-domain analyses that benchmark all three precision levels (FP32, FP16, and INT8) across different

medical datasets remain limited. This study seeks to fill that gap by systematically evaluating each quantization tier and offering an empirical basis for choosing the most balanced and reliable strategy for clinical AI deployment.

METHODOLOGY

This section outlines the workflow used in the study, including data preparation, model design, and the evaluation process for comparing the three quantization levels.

Data Selection and Preparation

To examine how quantization behaves across different clinical contexts, the study draws on three separate medical image collections: one for skin lesions, one for oral diseases, and one for nail disorders. Using datasets that represent distinct anatomical regions and visual patterns makes it possible to test whether the results generalize beyond a single type of pathology.

Domain	# of Classes	Image Count
Skin Lesions	10	27153
Oral Diseases	6	11653
Nail Disorders	6	3744

Table I
DATASET STATISTICS

Preprocessing Steps:

- **Resizing:** All images were resized to 224×224 pixels to match the input specification of the EfficientNet-Lite architecture.
- **Normalization:** Pixel values were scaled from the original 0–255 integer range to a 0.0–1.0 floating-point range, following standard preprocessing conventions for deep learning models.

Baseline Model Architecture and Training

The baseline system for all experiments is built on the EfficientNet-Lite0 architecture. This model was chosen because it offers a strong trade-off between computational cost and classification accuracy, making it well-suited for on-device inference. For each domain, the full-precision (FP32) Keras model is treated as the reference point against which all quantized variants are compared.

Training Protocol:

- **Transfer Learning:** Each domain-specific model was initialized with ImageNet pre-trained weights. These weights were then fine-tuned on the corresponding medical image dataset to adapt the network to the specialized visual patterns and class distributions.
- **Hyper-parameters:** Training was conducted using the Adam optimizer with an initial learning rate of 0.001. A batch size of 16 was used to accommodate GPU memory limitations. Model training was governed by an Early Stopping callback configured to monitor validation loss, resulting in convergence around epoch 87 (out

of a maximum of 100). Because each dataset involves multiple diagnostic categories, Categorical Cross-entropy was employed as the loss function.

Model Quantization Strategies

All optimized models were produced from the FP32 Keras baselines using the TensorFlow Lite (TFLite) Converter. For each domain, two additional versions were generated: an FP16 model and a fully quantized INT8 model.

16-bit Floating Point (FP16) Quantization: In this configuration, the model’s FP32 weights are cast to FP16. Because the arithmetic during inference remains in floating point, no calibration data is required. This technique mainly reduces memory consumption and file size—typically by about half—while maintaining performance that is nearly indistinguishable from the original FP32 model.

8-bit Integer (INT8) Quantization: To maximize deployment efficiency on mobile and embedded devices, full integer quantization was applied. In this process, both the weights and the intermediate activations are converted to 8-bit integers.

To ensure that the converted model actually runs with full integer acceleration, the TFLite Converter was configured to use the `TFLITE_BUILTINS_INT8` operator set, with both the input and output types forced to `tf.uint8`. This explicit configuration eliminates fallback to floating-point kernels, which can otherwise occur silently and undermine the benefits of quantization.

A small representative calibration set (100 randomly selected images) was used to estimate activation ranges. These examples help the converter map the original FP32 values into the more limited INT8 scale, reducing the risk of accuracy loss during inference.

Comparative Evaluation Protocol

A single evaluation procedure was applied to all nine models (three domains × three quantization tiers) to ensure that the reported efficiency and performance metrics are directly comparable. All measurements were taken on the same hardware and software configuration, preventing variations that could otherwise distort timing or resource-related results.

Evaluation Metrics:

- **Diagnostic Accuracy (%):** Accuracy was computed on each model’s fixed validation/test set. This measures how well the quantized versions retain the diagnostic reliability of their FP32 baselines.
- **Model Size (MB):** The size of each model file—whether stored as a Keras `.h5` or a TFLite `.tflite`—was taken directly from disk. This reflects the memory footprint and the storage or transmission requirements associated with deployment.
- **Inference Time (s):** Inference latency was estimated by running predictions on a batch of 100 test images and calculating the average time per image. This approach reduces noise and provides a stable indicator of deployment speed.

- **CPU Utilization (%):** During the inference runs, the average CPU usage was recorded. This metric helps characterize the computational load imposed by each quantization level and is particularly relevant for assessing heat generation, battery impact, and overall device responsiveness.
- **Runtime Memory Usage (MB):** Peak RAM consumption was measured while loading each model and performing inference. This provides a practical view of memory demands in resource-limited environments and indicates how efficiently the model can coexist with other processes on the device.

RESULTS AND DISCUSSION: PERFORMANCE OPTIMIZATION AND CLASSIFICATION ANALYSIS

The following section contains the results of the performed quantization research. Efficiency (model size, latency) are the main metrics reported, and additional context is provided about the classification accuracy in terms of the restrictions imposed by the methodological approach of this work.

Performance Gains: Latency, Compression, and Resource Efficiency

The main hypothesis of the work is that quantization methods can be successfully leveraged to compress and accelerate the original EfficientNet Lite architecture for the target low-resource mobile environment. The results obtained herein prove this hypothesis beyond doubt: impressive gains in all the main performance indicators are observed.

Model Version	Size (MB)	Latency (s)	Avg. CPU (%)	Max. RAM (MB)
FP32 (Baseline)	46.96	0.166	58.96	1183
FP16	7.72	0.0275	49.71	1046
8-Bit Int (Quant.)	4.69	0.0173	50.08	911

Table II
MODEL CHARACTERISTICS

Model Version	Compression Factor	Speed Increase (vs. FP32)
FP16	6x	6x
8-Bit Integer (Quantized)	10x	9.5x

Table III
PERFORMANCE METRICS

- **Model Compression:** Full 8-bit integer quantization successfully reduced the model size from 46.96 MB to just 4.69 MB, achieving a 10x compression factor. This outcome is crucial for minimizing application download size and device storage requirements.
- **Inference Speed:** The optimized 8-bit model achieved an average inference latency of only 17 milliseconds per image. This represents an 9.5x speed increase compared to the FP32 baseline, validating the model’s suitability for real-time execution on resource-constrained platforms.

- **Resource Efficiency (RAM and CPU):** The quantization process led to a significant reduction in runtime memory footprint. Maximum RAM usage dropped from 1183.76 MB (FP32 baseline) to 911.69 MB for the 8-bit model, enhancing system stability. While average CPU utilization remained constant, the dramatic reduction in execution time translates directly into a substantial gain in energy efficiency per inference.

Analysis of Classification Accuracy and Methodological Limitations

The resulting classification accuracy requires interpretation based on the source model's inherent limitations in generalization.

The model achieved a high accuracy of approximately 90% on the training dataset. However, its performance on real-world test data reflecting the combined domain of Skin, Oral, and Nail images already dropped significantly to approximately 30% due to the heterogeneous nature and restricted volume of the dataset. Consequently, the model's potential for robust classification was severely constrained by this initial lack of generalizability, suggesting that the bottleneck lies within the source data's variance.

In summary, the classification accuracy scores ($\approx 30\%$) reflect the model's intrinsic weakness in generalization—a limitation stemming from the limited and heterogeneous training data—and not a primary failure of the quantization methodology itself. The critical metric for deployment, efficiency, was achieved without compromise. The 10x compression and 9.5x acceleration unequivocally demonstrate that the core objective of the research—maximizing the portability and inference speed of the EfficientNet Lite architecture—was successfully validated. Thus, the evidence confirms the efficacy of 8-bit quantization as an essential strategy for achieving real-time performance on resource-constrained platforms.

CONCLUSION

This study set out to examine how different quantization levels affect the behavior of an EfficientNet Lite model when deployed on devices with tight computational limits. By evaluating the three formats—FP32, FP16, and INT8—across datasets from skin, oral, and nail imaging, the work provides a clearer sense of what can realistically be expected when these models are pushed toward mobile or embedded settings.

Across all experiments, the INT8 version stood out. The jump in efficiency was substantial: the model shrank from roughly 46 MB to just under 5 MB, and the time needed for a single prediction fell to about 17 milliseconds. Those gains, taken together, show that full integer quantization is not just a convenient option but the most practical route for real-time inference on edge hardware. The reduced memory usage and lighter CPU load further reinforce this, especially for battery-powered devices that cannot afford sustained high computational demand.

The lower accuracy observed on the test set—hovering around thirty percent—should be viewed in the context of the model's earlier limitations. The original dataset was both small and highly varied, and the baseline model already struggled to generalize before any optimization was applied. Quantization did not introduce this gap; it simply made an existing issue more visible. Even so, the efficiency improvements demonstrate that the main deployment goal was met.

In short, the results point to INT8 quantization as the most effective choice for situations where speed, size, and resource use matter as much as accuracy. Looking ahead, the natural next step is to work with a larger and more coherent dataset. A stronger foundation at the data level would allow these optimized models to reach the diagnostic performance they are theoretically capable of achieving.

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